STATEMENT OF WORK FOR
AIRBORNE SYSTEMS INTEGRATION DIVISION
SECURE EXECUTIVE NETWORK SOFTWARE SUPPORT

1.0 Scope.

As the organic Lead Systems Integrator® (oLSI®) for the Naval Air Warfare Center Aircraft Division (NAWCAD), Webster Outlying Field (WOLF) operates under a product-based business model whereby NAWCAD WOLF civilians lead and manage the execution of the requirements with combined Government-Industry teams. Using this oLSI® approach, the Government maintains control of intellectual property and data rights, enabling WOLF to rapidly adapt solutions to the dynamic needs of the warfighter.

The NAWCAD WOLF Airborne Systems Integration (ASI) Division provides the rapid insertion and transition of advanced technologies, systems, sensors, and weapons which enable Programs of Record (PoR) to successfully execute and deliver the direct insertion of capabilities to meet urgent fleet needs. The NAWCAD WOLF ASI Division, operating under the oLSI® model, has requirements for engineering and integration of various airborne capabilities to include, but not be limited to, sensors, communications systems, weapons systems, and control technologies for a variety of manned and unmanned airborne platforms. Specifically, the ASI Division requires the Contractor to support development, design, prototyping, low-rate production, demonstration, testing, integration, technical installation, transition through Initial Operational Capability (IOC), and maintenance in the domains of cyber, Maritime Domain Awareness (MDA), situational awareness, data management and dissemination, and other rapid capability software requirements with the ASI Division in support of Department of Navy (DON), Department of Defense (DoD), and other Government agency projects. This SOW includes, but is not limited to, software development, network design and architecture, cyber engineering, cybersecurity accreditation support, engineering services, installation, integration, design engineering for airborne capabilities, test and evaluation, and user and maintainer training.

2.0 Applicable Documents.

The documents listed below are presented only as representative sources of the technological interface details that may be required during TO performance. Likewise, the documentation listed is not all-inclusive, but is representative of the type of information that may be necessary to perform the work. The applicable document version is the latest version at time of award.

2.1 Department of Defense (DoD) Specifications.

2.1.1 DoD National Industrial Security Program Operating Manual (NISPOM) codifying 32 Code of Federal Regulation Part 117, NISPOM Rule.

2.1.2 DoDM 5200.01 Vol. 2, Marking of Classified Information, Change 4, 28 Jul 2020.

2.1.3 SECNAV M-5510.36B, Department of the Navy, Information Security Program, 12 Jul 2019.

2.1.4 DoDI 5200.48, Controlled Unclassified Information (CUI), 6 Mar 2020.

2.1.5 DoDD 5400.07, DoD Freedom of Information Act (FOIA) Program, 5 Apr 2019.

*2.1.6 DoDI 5230.24, Distribution Statements on Technical Documents, Change 3,*10 Jan 2023.

2.1.7 SECNAV M-5510.30C, Department of the Navy, Personnel Security Program, 24 Jan 2020.

2.1.8 OPNAVINST 3440.17A, Navy Installation Emergency Management Program, 1 Aug 2014.

2.1.9 DoDI 1035.01, Telework and Remote Work, 8 Jan 2024.

2.1.10 SECNAVINST 5239.3B, Department of the Navy Information Assurance Policy.

2.1.11 SECNAVINST 5239.25, Department of the Navy Cyberspace Workforce Qualification and Management Program, 10 Oct 2023.

*2.1.12 DoDD 8140.01,Cyberspace Workforce Management,*5 Oct 2020.

2.1.13 DOD Manual 8140.03 Cyberspace Workforce Qualification and Management Program, 15 Feb 2023.

2.2 DoD Standards.

2.2.1 DoDI 8510.01, Risk Management Framework (RMF) for DoD Systems, 19 Jul 2022.

2.2.2. DoDI 8500.01, Cybersecurity.

2.2.3 DoDM 4161.2-M, DoD Manual for the Performance of Contract Property Administration.

2.2.4 DoDM 5200.02, Procedures for the DoD Personnel Security Program (PSP).

2.2.5 DoDI 8140.02, Identification, Tracking, and Reporting of Cyberspace Workforce Requirements.

2.3 Other Government Documents.

2.3.1 National Institute of Standards and Technology (NIST) Special Publication (SP), 800-59.

2.3.2 National Security Presidential Memorandum-28 (NSPM-28), dated 05 Jun 1999.

2.3.3 NAVAIRINST 4355.19E, Systems Engineering Technical Review Process.

2.3.4 Intelligence Community Directive (ICD) 503, Intelligence Community Information Environment Risk Management, dated Oct 2024.

2.3.5 NAVAIR SWP4P00-047.3, Cybersafe Certification.

2.4 Industry documents.

2.4.1 American Society for Testing and Materials (ASTM) Manual A.

2.4.2 Institute of Electrical and Electronics Engineers (IEEE) 12207-2017 – ISO/IEC/IEEE International Standard – Systems and software engineering – Software life cycle processes.

3.0 Requirements.

3.1 General Requirements.

3.1.1 Compatibility. The Contractor shall maintain the capability to prepare documents and software packages compatible with the Government Information Technology (IT) environment through the security classification of Secret as specified on DD Form 254, Contract Security Classification Specification Form (Attachment 01). The current operating environment required for this TO includes:

Microsoft Windows 2016 or greater

Microsoft Project 2016 or greater

Microsoft Office Professional Plus 2016 or greater

Adobe Acrobat Reader

Internet access

Flank Speed

Microsoft Office 365

The Contractor shall maintain the ability to interface with and transfer data to and from requiring office software applications and their upgraded versions. The Contractor shall ensure that all media are virus free when delivered. The Contractor shall be capable of internet and Local Area Network (LAN) communications with the ASI Division. Contractor personnel shall be capable of maintaining real-time communications, both voice and data transfer capabilities, with the ASI Division during working hours whether at Contractor work site or on travel.

3.1.2 Work Location, Facilities, and Telework.

3.1.2.1 Work Location. Approximately 51% of work will be performed at Government site and 49% of work to be performed at contractor site. Government sites include NAWCAD WOLF, St. Inigoes, MD and Naval Air Station (NAS) Patuxent River, Patuxent River, MD.

3.1.2.1.1 Government Site Requirements. Contractors performing on-site support will be provided access to workspaces, telephones, printers, facsimile machines, copy machines, shredders, computers, and network access including web servers and applicable databases or other applications necessary to carry out assigned tasks.

3.1.2.1.2 Contractor Site Requirements. The Contractor shall provide its own primary support facilities needed to perform the tasking of this TO, which must be located within 50 drivable miles of NAWCAD WOLF, St. Inigoes, MD in order to satisfy the daily support and administrative operations of this TO. The cost of general purpose business items/expenses required for the Contractor to conduct its normal business operations will not be considered an allowable direct cost in the performance of this TO. General purpose business items include, but are not limited to, the cost for items such as telephones and telephone charges, reproduction machines, personal computers, and other office equipment and supplies. The Government will not assume responsibility to retain the facility or take control of the title after the TO is complete.

3.1.2.2 Meeting Support. In support of the tasking outlined in this Statement of Work (SOW), the Contractor shall have the capability to host and conduct meetings at the classification level up to secret with the capacity to support a minimum of 15 persons. This support shall be provided within 50 drivable miles of NAWCAD WOLF, St. Inigoes, MD.

3.1.2.3 Telework.

The Contractor may allow Contractor employees to telework in accordance with (IAW) DoDI 1035.01, Telework and Remote Work, and in section 3.4 of this SOW. The Contractor may utilize alternate work sites/locations to telework to support continued performance of SOW tasks IAW company policy. When using telework, the Contractor remains responsible for TO performance and compliance with any applicable cost accounting standards and contract cost principles/procedures.

Telework is defined as an alternative workplace arrangement under which a position performs the duties and responsibilities of the assigned position and other authorized activities from an approved alternative worksite but is required to report to the agency's worksite on a regular and recurring basis. Telework can be either routine or situational. Routine telework is performed from an alternative worksite on regularly scheduled telework day(s), whereas situational telework is authorized on a case-by-case basis as the need arises or when opting for unscheduled telework and may be authorized for routine teleworkers in such circumstances on regularly scheduled in-office days.

Telework-eligible positions require routine on-site performance of two days per week. On-site performance may be required outside of routine on-site performance days and shall be provided when requested by the Contracting Officer's Representative (COR).

At any time during TO performance, the COR may determine that a position is no longer eligible for routine telework and/or situational telework by providing a notice 30 calendar days in advance so the Contractor can address changes needed to support mission requirements.

3.1.3 TO Status Reporting. The Contractor shall provide the following documentation.

3.1.3.1 Monthly Progress and Financial Status Report. The Contractor shall provide a progress and financial status report IAW the Contract Data Requirements List (CDRL). The report shall include work accomplished since submittal of the last report, both monthly and cumulative work hour labor costs expended by labor category, and material and travel costs. (CDRL A001)

3.1.4 Work schedule to include Compressed Work Schedule (CWS), holidays, and installation closure.

3.1.4.1 Work Schedule. The Contractor shall provide the required services and staffing coverage during normal working hours. Normal working hours are usually 8.5 hours (including a 30-minute lunch break), from 0730 to 1600 each Monday through Friday (except on the legal holidays as specified in paragraph 3.1.4.1.2). Some supported Government offices have flexibility to start as early as 0600 and end as late as 1800 Monday – Friday.

3.1.4.1.1 CWS. CWS is an alternative work schedule to the traditional 5 8.5-hour workdays (which includes a 30-minute lunch break) worked per week. Under a CWS schedule, an employee completes the following schedule within a 2-week period of time: 8 weekdays are worked at 9.5 hours each (which includes a 30-minute lunch break), 1 weekday is alternately worked at 8.5 hours (which includes a 30-minute lunch break), and 1 weekday is not worked by the employee. The result is 80 hours worked every 2 weeks, with 44 work hours 1 week and 36 work hours the other.

The Contractor may allow its employees to work a CWS schedule provided the requirements of this SOW are met. If the Contractor chooses to allow its employees to work a CWS schedule in support of this TO, any additional costs associated with the implementation of the CWS schedule vice the standard schedule are unallowable costs under this TO and will not be reimbursed by the Government. Additionally, the CWS schedule shall not prevent contractor employees from providing necessary staffing and services coverage as required by the Government to the COR/Alternate Contracting Officer's Representative (ACOR).

3.1.4.1.2 Holidays. The Government observes the Federal holidays identified on the Office of Personnel Management (OPM) website: https://www.opm.gov/policy-data-oversight/pay-leave/federal-holidays/#url=Overview. With the exception of the events in section 3.1.4.1.3 below, the Contractor is permitted to observe Federal holidays IAW its corporate policy.

3.1.4.1.3 Installation Closure. When Federal facilities are closed by the Government or when Federal employees are officially excused from work due to a holiday or a special event, severe weather, a security threat, or any other Government facility-related problem that prevents Federal personnel from working at the Government facility, Contractor personnel assigned to work at that facility in support of such Federal employees shall follow their parent company's policies.

While generally Contractor personnel may not perform work on-site at a Government facility without oversight from Federal personnel, in very limited circumstances, work being performed by contractor personnel may be deemed mission essential and performance of such mission essential work may be authorized to continue at the Government facility despite the facility being otherwise closed for normal operations. The circumstances permitting work being performed by Contractor personnel to be deemed mission essential are extremely limited and generally only apply to performance of efforts related to public health, safety, or matters related to national security shall be IAW National Security Decision Directive 298. The cognizant Contracting Officer (KO) must concur with any determination that work being performed by Contractor personnel is mission essential.

3.1.5 Other Direct Costs (ODCs). Procurement (PROC); Operations and Maintenance (O&M); Foreign Military Sales (FMS); Research, Development, Test and Evaluation (RDT&E); and Working Capital Funds (WCF).

3.1.5.1 Travel. Travel may include general and administrative expenses but shall not include profit. Temporary travel to other locations in support of program tasking is required. If required, temporary travel includes both Continental United States (CONUS) and Outside the Continental United States (OCONUS) locations. CONUS locations may include, but are not limited to: San Diego, CA; Jacksonville, FL; Lake Charles, LA; Oklahoma City, OK; Waco, TX; and Norfolk, VA. OCONUS locations may include, but are not limited to: Australia, Guam, Japan, and Korea. This list is not all inclusive as locations may change over the life of the TO.

3.1.5.1.1 Local Travel Reimbursement. Reimbursement of local travel to and from NAS Patuxent River, NAWCAD WOLF, the Contractor's facility or facilities, and the surrounding area by the Contractor or Subcontractor at a distance equal to or less than 50 drivable miles of NAWCAD WOLF is not an allowable cost.

3.1.5.1.2 Travel Request Authorization. The Contractor shall complete travel requests IAW *CTXT.232-9509, Travel Approval and Reimbursement Procedures*. Upon completion of the travel, the Contractor shall complete a travel report. (CDRL A002)

3.1.5.1.3 Synchronized Pre-Deployment & Operational Tracker (SPOT). The Contractor may travel to U.S. Central Command locations. IAW *Clauses 252.225-7040* and *5152.225-5908, SPOT* enables the validation of Contractors Authorized to Accompany the Force (CAAF), their authorization and eligibility for access to specific DoD facilities, and their eligibility for specific Government Furnished Support (GFS). The Contractor shall initiate a Letter of Authorization (LOA) for each prospective traveler. The Contractor shall use the SPOT link (https://spot.dmdc.mil/privacy.aspx) to enter and maintain data with respect to traveling/deployed personnel and to generate LOAs.

3.1.5.1.4 LOA. The Contracting Officer will provide an LOA for official travel OCONUS, when applicable, in support of the TO. The LOA will identify local authorizations, privileges, etc. as specified by DoD requirements. All defense contractor employees working under this TO shall carry an LOA with them, when applicable, at all times while deployed OCONUS.

3.1.5.2 Material. Incidental material will be required in the performance of this TO, and all incidental material purchases shall be IAW *CTXT-242.9520, Procedures and Approvals Required Prior to Incurring Direct Material Costs*. All materials not depleted during the performance of this TO shall become Government property upon completion of this TO. The Contractor shall transfer all materials not depleted to the COR by way of a *DD Form 250, Material Inspection and Receiving Report*. Material costs may include general and administrative expenses but shall not include profit/fee. (CDRL A003)

3.1.6 Subcontractors. Provisions stated herein shall be clearly and effectively communicated to all subcontractors providing support under this TO. All provisions of this SOW shall flow down to subcontractors providing support under this TO.

3.1.6.1 The Contractor shall submit a Small Business Participation Commitment Document (SBPCD) (CDRL A004).

3.1.7 Management of Contractor Personnel. The Government will neither supervise Contractor employees nor control the method by which the Contractor performs the required tasks. Under no circumstances will the Government assign tasks to, or prepare work schedules for, individual Contractor employees. The Contractor shall manage its employees and guard against any actions that are of the nature of personal services or actions that give the perception of personal services.

3.1.8 Transition-Out Strategy.

The Contractor's overall transition-out strategy shall be built around maintaining the mission of the ASI Division with minimal impact, not only in terms of timeliness of performance but also to ensure that critical data and knowledge transfer occurs. Prior to the termination or expiration of the TO, the Contractor shall ensure an orderly transition of responsibilities, while minimizing impact to the operation. The Contractor shall submit a Transition-Out Plan, to include the minimum elements listed below IAW CDRL A005.

- Work turnover. The Contractor shall provide a plan of action to effectively transfer tasked work that is in process at the expiration or termination of the TO to the successor company and establish and maintain effective communication with the incoming Contractor or Government personnel for the period of transition via weekly status meetings.

- Quality Assurance (QA). The Contractor shall provide a plan of action to ensure continuation of quality review processes during the transition period to the successor company.

- Risk mitigation strategies. The Contractor shall provide a plan of action to mitigate TO performance risks (quality and schedule) encountered during the transition period.

- Data/information transfer. The Contractor shall provide a plan of action for the efficient inventory and transfer of program data to the successor company.

3.1.9 Technical Direction Letters (TDLs). When necessary, technical direction or clarification concerning the details of specific tasks set forth in the TO will be given through issuance of written TDLs. TDLs will not, in any manner, alter the scope of the TO. For further direction see *HTXT.242-9502, Technical Direction (Variation)* in Section H of the TO. The Contractor shall prepare and deliver a *COR Management Report for TDLs* IAW CDRL A001.

3.1.10 Organizational Conflicts of Interest (OCI) and Non-Disclosure Agreement (NDA) Status Certification. The Contractor shall report *OCI Mitigation Plan* compliance and compliance status changes. (CDRL A006)

3.1.11 Government Furnished Property (GFP). Any GFP that may be required will be listed during execution of the TO and will be provided to the Contractor to support the TO. Disposition of GFP will be made at TO completion. (CDRLs A007)

3.2 Security.

3.2.1 Citizenship Requirements. Only U.S. citizens may perform under this TO, unless waived by the Government. If the Contractor cannot find qualified U.S. citizens to perform the work, the Contractor shall submit a citizenship waiver request with justification to the Government Security Office. The waiver request should include:

a) The individual's name, date and place of birth, position title, and current citizenship.

b) A statement that a qualified U.S. citizen cannot be hired in sufficient time to meet the contractual requirements.

c) A statement of the unusual expertise possessed by the applicant.

d) A statement that access will be limited to a specific Government contract (specify contract number).

e) A statement that the Contractor has obtained an export license for the information required to perform the TO.

3.2.2 Investigative Requirements.

Unclassified: All Contractor personnel must be eligible to perform Non-Critical Sensitive work as defined by *SECNAV M-5510.30C*. All Contractor personnel are required to have a favorably adjudicated Tier-3 investigation from the OPM. The Contractor shall submit a request for personnel security investigation to the Government Security Office. The Government Security Office shall initiate the Contractor employee's Electronic Questionnaire for Investigations Processing (eQIP) and shall perform a preliminary screening of the Contractor employee's eQIP for suitability and derogatory information. The Contractor employee shall provide all requested information pursuant to the *Privacy Act of 1974*. The Government Security Office may deny the Contractor employee access to Government facilities and information and may prohibit the Contractor employee from performing sensitive duties for failure to provide requested information or when derogatory or adverse information is present on the Contractor employee's eQIP. In such cases, the Contractor employee may not perform on the TO.

Classified: All Contractor personnel shall maintain security clearance eligibility commensurate with the level of classification of the work performed as annotated in the Contract's *DD Form 254, Contract Security Classification Specification Form*. Contractor personnel shall require access to classified information in performance of this TO up to and including Top Secret Special Compartmentalized Information (TS SCI) facility level per *Director of Central Intelligence Directive (DCID) 6/3, Protecting Sensitive Compartmented Information within Information Systems*, with a safeguarding level of Secret. Contractor personnel will require access to classified information in performance of this TO up to and including Top Secret (TS)/Sensitive Compartmented Information (SCI). The Contractor will require a final Top Secret Facility Clearance Level (FCL), with an approved safeguarding level of Secret.

The Contractor is responsible for ensuring that all personnel receive the requisite investigation and are favorably adjudicated IAW *National Industrial Security Program Operating Manual (NISPOM) codifying 32 Code of Federal Regulation Part 117, NISPOM Rule* and *DoDM 5200.01 Vol. 2, Marking of Classified Information, Change 4,* 28 Jul 2020. Contractor employees who fail to meet security clearance requirements may not access classified information or perform sensitive duties. In such cases, the Contractor employee may not perform on the TO.

3.2.3 Contractor Personnel Labor Category Security Requirements. All personnel shall have, at a minimum, interim Secret clearances within 60 calendar days of award with the exceptions noted in the table below.

LABOR CATEGORY	DoD SECURITY CLEARANCE LEVEL (INTERIM CLEARANCE ACCEPTABLE)	NUMBER OF CLEARED EMPLOYEES REQUIRED	OFF-SITE/ON-SITE	REQUIRED CALENDAR DAYS AFTER ISSUANCE OF TO
Computer Network Architects, Intermediate, 15-1241	Top Secret with favorable Tier 5	1	On-Site	90
Computer Systems Analysts, Advanced, 15-1211	Top Secret with favorable Tier 5	1	On-Site	90
Engineers, All Other (Systems Engineer), Journeyman, 17-2199	Top Secret with favorable Tier 5	1	On-Site	90
Engineers, All Other (Systems Engineer), Senior, 17-2199, KEY	Top Secret with favorable Tier 5	1	Off-Site	90
General and Operations Managers (Tactical Network and Computer Systems Administrator), Intermediate, 11-1021	Top Secret with favorable Tier 5	1	On-Site	90

General and Operations Managers (Tactical Network and Computer Systems Architect), Advanced, 11-1021	Top Secret with favorable Tier 5	1	Off-Site	90
Information Security Analysts, Advanced, 15-1212	Top Secret with favorable Tier 5	1	On-Site	90
Network and Computer Systems Administrators, Intermediate, 15-1244	Top Secret with favorable Tier 5	1	Off-Site	90
Operations Research Analysts (Cyber Analyst/Modeler), Advanced, 15-2031	Top Secret with favorable Tier 5	1	Off-Site	90
Software Developers, Intermediate, 15-1252	Top Secret with favorable Tier 5	2	On-Site	90
Software Developers, Advanced, 15-1252, KEY	Top Secret with favorable Tier 5	1	On-Site	90
Software Quality Assurance Analysts and Testers, Advanced, 15-1253	Top Secret with favorable Tier 5	1	On-Site	90

3.2.4 Common Access Card (CAC)/Public Key Infrastructure (PKI) and System Authorization Access Request (SAAR).

3.2.4.1 SAAR. All Contractor personnel requiring access to Government IT systems shall have an approved *DD Form 2875, System Authorization Access Request (SAAR)* form on file and complete required *Annual Information Awareness Training*. New employees must submit their SAAR forms within 30 business days of their first day of work. SAAR forms shall be submitted to the COR, Government Technical Point of Contact (TPOC), or to the assigned Government Mission Partner Identity, Credential, and Access Management (MP ICAM) Trusted Associate (TA).

3.2.4.2 CAC/Local Badges. Contractor CACs and facility-specific identification badges will be issued by the Government to on-site Contractor personnel and shall be visible at all times while personnel are at the Government site. The Contractor shall furnish all requested information required to facilitate issuance of identification badges. All CACs and identification badges issued to Contractor employees shall be returned to the TA following completion of the TO, relocation or termination of an employee, or upon request from the COR/Procuring Contracting Officer (PCO). The Government will provide the Contractor access to Government facilities, as required, for performance of tasks under this TO.

3.2.4.3 DD Form 254. The Contractor shall comply with security requirements specified in the *DD Form 254* attached to this TO. Information or data that the Contractor accesses shall be handled at the appropriate classification level. Unclassified information shall be handled in accordance with the appropriate designation (Controlled Unclassified Information (CUI); Legacy

For Official Use Only (FOUO); Covered Defense Information (CDI)). Distribution is authorized to the Requiring Office's Organization and supported Activity only. Other requests for deliverables under this TO shall be referred to the TPOC/COR of this TO for approval.

3.2.5 Information Security. If the work is performed at the Contractor's facility, the Contractor shall implement and maintain security procedures and controls to prevent unauthorized disclosure of classified information and CUI and to control distribution of CUI IAW *National Industrial Security Program Operating Manual (NISPOM) codifying 32 Code of Federal Regulations Part 117, NISPOM Rule* and *SECNAV M-5510.36B, Department of the Navy, Information Security Program*. If the work is performed at the Government's facility, the Contractor shall comply with facility security procedures and controls.

All Contractor facilities shall provide an appropriate means of storage for CUI and materials. All CUI including legacy FOUO and CDI (meeting the definition *DFARS Clause 252.204–7012, Safeguarding Covered Defense Information and Cyber Incident Reporting*) generated and/or provided under this TO shall be marked and safeguarded as specified in *DoD Instruction 5200.48, Controlled Unclassified Information (CUI)* available at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520048p.PDF.

Any product containing CDI shall be assigned a distribution statement (distribution statements B through F) using the criteria set forth in *DoDI 5230.24, Distribution Statements on Technical Documents* and have this statement displayed per *DoDI 5230.24, Enclosures 3 and 4*. Distribution is authorized to the Requiring Office's Organization and supported Activity only. Other requests for deliverables under this TO shall be referred to the TPOC/COR of this TO for approval.

All controlled unclassified technical information shall be appropriately identified and marked with the following distribution statement:

Distribution Statement (Insert Appropriate Letter and Authorization Title), (Insert Appropriate Reason Category) (dated – (Date of Distribution Authorization)). Other requests shall be referred to: Commander, Naval Air Systems Command, Attn: ASI COR, 17800 Moll's Cove Road, Building 8125, St. Inigoes, MD 20684.

The Contract deliverables (including reports, software and hardware) will not be classified at the SCI level. SCI shall be utilized and accessed as part of routine work only.

3.2.5.1 Marking. All information generated by the Contractor shall be properly marked. Classified information shall be marked IAW *DoDM 5200.01, Vol 2, Marking of Classified Information*. CUI, including legacy FOUO information generated and/or provided under this TO, shall be marked IAW *DoDI 5200.48, Controlled Unclassified Information (CUI)*. Technical information shall also be marked with appropriate Distribution Statements and Export Control warnings IAW *DoDI 5230.24, Distribution Statements on Technical Documents* and program Security Classification Guidance.

3.2.5.2 Public Release. Any CUI pertaining to this TO shall not be released for public dissemination, including posting to any social media sites such as Facebook or X (formerly known as Twitter), unless it has been approved for public release by appropriate U.S. Government authority. Proposed public releases shall be submitted for approval prior to release through the NAWCAD WOLF ASI Division COR, *NAVAIR Form 5720.10*, which must be completed and attached.

3.2.5.3 Loss, compromise, and/or electronic spillage of classified information or CUI. All instances of loss, compromise, and electronic spillage of classified information or CUI shall be reported to the COR, TPOC, and Government Security Office within 72 hours of the incident occurring.

3.2.6 Operations Security (OPSEC). The Contractor shall develop, implement, and maintain an OPSEC program to protect controlled unclassified and classified activities, information, equipment, and material used or developed by the Contractor and any Subcontractor during performance of the TO. The Contractor shall be responsible for the subcontractor implementation of the OPSEC requirements. This program may include Information Assurance (IA) and Communications Security (COMSEC). The OPSEC program shall be IAW *National Security Decision Directive (NSDD) 298* and at a minimum shall include:

1) Assignment of responsibility for OPSEC direction and implementation.

2) Issuance of procedures and planning guidance for the use of OPSEC techniques to identify vulnerabilities and apply applicable countermeasures.

3) Establishment of OPSEC education and awareness training.

4) Provisions for management, annual review, and evaluation of OPSEC programs.

5) Flow down of OPSEC requirements to subcontractors when applicable.

While performing at Naval Air Systems Command (NAVAIR) or NAVAIR sites, the Contractor shall comply with facility OPSEC program instructions and contribute to organization-level OPSEC efforts; include OPSEC as part of its ongoing security awareness program and take all required Agency training; be responsive to the Supporting OPSEC Manager on a non-interference basis; and protect sensitive unclassified information and activities that could compromise classified information or operations or degrade the planning and execution of operations performed by the Requirements Owner and Contractor in support of the mission. (CDRL A008)

3.2.7 Antiterrorism Force Protection and Emergency Management. The work performed on this TO is not Emergency Essential IAW *OPNAVINST 3440.17A, Navy Installation Emergency Management Program* and Government Emergency Management, Antiterrorism, and/or Continuity of Operations Plans. Contractor personnel shall comply with all Government Emergency Management, Antiterrorism, and/or Continuity of Operations Plans and directives.

Contractor personnel shall not report for work at Government facilities upon declaration of Force Protection Condition CHARLIE or in any event or emergency where Government officials direct curtailment of operations to "Mission Essential Only." All Contractor personnel assigned to a Government facility shall complete annual *Antiterrorism (Level One)* and *Active Shooter* training.

3.3 Detailed Support Requirements.

3.3.1 Software Support.

3.3.1(a) (PROC) Provide software support to modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks during the production phase of a project's life cycle. Applicable to paragraphs 3.3.1.1 – 3.3.1.3.

3.3.1(b) (O&M) Provide software support to operate, maintain, refurbish, overhaul, train, and perform in-service engineering on software, computers, and networks. Applicable to paragraphs 3.3.1.1 – 3.3.1.3.

3.3.1(c) (FMS) Provide software support to develop, prototype, evaluate, establish performance capabilities, operate, maintain, refurbish, overhaul, train, perform in-service engineering, modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks for FMS customers. Applicable to paragraphs 3.3.1.1 – 3.3.1.3.

3.3.1(d) (RDT&E) Provide software support to research, develop, prototype, evaluate, establish performance capabilities, support associated testing and evaluation efforts, and support training in order to meet developmental milestones prior to operational use for software, computers, and networks during the RDT&E phase of a project's life cycle. Applicable to paragraphs 3.3.1.1 – 3.3.1.3.

3.3.1(e) (WCF) Provide software support to develop, prototype, evaluate, establish performance capabilities, operate, maintain, refurbish, overhaul, train, perform in-service engineering, modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks during all phases of a project's life cycle IAW *Institute of Electrical and Electronics Engineers (IEEE) 12207-2017 – ISO/IEC/IEEE International Standard – Systems and software engineering – Software life cycle processes.* Applicable to paragraphs 3.3.1.1 – 3.3.1.3.

3.3.1.1 Software Development Support.

3.3.1.1.1 The Contractor shall support the analysis of requirements for and development of software to include coding and implementation according to applicable coding standards (e.g., the *Institute of Electrical and Electronics Engineers (IEEE) 12207-2017 – ISO/IEC/IEEE International Standard – Systems and software engineering – Software life cycle processes*), Security Technical Implementation Guides (STIGs), and software governance models. (CDRL A009)

3.3.1.1.2 The Contractor shall provide support by researching, analyzing, developing, and modifying software architectures to meet system requirements and by documenting the architectures in engineering documentation including software requirements specifications, software description documentation, software test plans, and requirements verification matrices. (CDRLs A010, A011, A012, and A013)

3.3.1.1.3 The Contractor shall support the development of software needed to ensure that aircraft mission systems, ground support equipment, and operator systems are operational and tactically relevant. This may include, but is not limited to, system integration support and fusion of multiple sensors and technologies, communication devices, data sources, and disparate networks. Representative platform sensors may include, but not be limited to, Geospatial Intelligence (GEOINT), Imagery Intelligence (IMINT), Signals Intelligence (SIGINT), Measurement and Signature Intelligence (MASINT), imagery, and communications. Representative communication systems may include, but not be limited to, Link 16, Digital Video Broadcast Second Generation/Extended (DVB-S2/X), Multi-User Object System (MUOS), and Commercial-Off-the-Shelf (COTS) and Government-Off-the-Shelf (GOTS) Mobile Ad-Hoc Networking (MANET).

3.3.1.1.4 The Contractor shall support the development of software needed to record and analyze data from system, communication, and sensor technologies. (CDRL A009)

3.3.1.1.5 The Contractor shall support the development of embedded system design software to include, but not be limited to, Programmatic Logic Controllers (PLCs) and Microprocessors using assembly level language as well as higher level languages to include, but not be limited to, Ada, C++ and Java. (CDRL A009)

3.3.1.1.6 The Contractor shall support the development of real time application software for weapon systems, platforms, and networks. Software may be mission critical, flight safety critical, and/or National Defense time critical. (CDRL A009)

3.3.1.1.7 The Contractor shall support the development of bi-directional custom control software and/or firmware to interface with mission system components and IT infrastructure. (CDRL A009)

3.3.1.1.8 The Contractor shall support the development of application software for information systems that support the development, tracking, monitoring, analysis, sustainment, situational awareness, decision making, data fusion, data reduction, and data visualization for DoD business and mission systems. (CDRL A009)

3.3.1.1.9 The Contractor shall support the creation and maintenance of a software repository for COTS, GOTS, and Government-purpose software to include technical documentation, system configurations, and source code. (CDRL A014)

3.3.1.1.10 The Contractor shall support the development of algorithms, autonomy, machine learning, artificial intelligence, and deep learning for advanced selection and processing techniques to enable new cyber capabilities or enhancements to existing capabilities to include, but not be limited to, signal processing, data processing, sensor calibration, performance assessment, and real-time and post-processing tools.

3.3.1.1.11 The Contractor shall support the design and development of software security constructs to include, but not be limited, to Multi-Level Security (MLS) interfaces, devices, and rule creation.

3.3.1.1.12 The Contractor shall support the design and development of prototype software to evaluate and verify the inclusion of capabilities into software baselines.

3.3.1.1.13 The Contractor shall support the research and development of software to consolidate, evaluate, and provide the user with fault detection, isolation, and resolution procedures for the system.

3.3.1.1.14 The Contractor shall support researching and developing intuitive user interfaces and control of complex software implementations in streamlined applications.

3.3.1.2 Software Documentation Support.

3.3.1.2.1 The Contractor shall support the creation of necessary software documentation and artifacts related to the development, use, and administration of software IAW applicable coding guidelines (e.g., *Institute of Electrical and Electronics Engineers (IEEE) 12207-2017 – ISO/IEC/IEEE International Standard – Systems and software engineering – Software life cycle processes* and Capability Maturity Model Integration (CMMI) guidelines).

3.3.1.2.2 The Contractor shall support the Government with the development of System Engineering Technical Review (SETR) documentation as it relates to the development of software solutions. IAW *NAVAIRINST 4355.19E, Systems Engineering Technical Review Process.*

3.3.1.2.3 The Contractor shall support the development of Software Requirement Specifications. (CDRL A010)

3.3.1.2.4 The Contractor shall support analyzing and updating configuration control of software baselines to include tracking changes between major, minor, and incremental software builds in Software Design Descriptions (SDDs). (CDRL A011)

3.3.1.2.5 The Contractor shall support the development of Software Version Descriptions (SVDs). (CDRL A015)

3.3.1.2.6 The Contractor shall support the development of Interface Requirements Specifications. (CDRL A016)

3.3.1.2.7 The Contractor shall support the development of Interface Design Descriptions (IDDs). (CDRL A017)

3.3.1.2.8 The Contractor shall support the development of System/Subsystem Design Descriptions. (CDRL A018)

3.3.1.2.9 The Contractor shall support the development of verification matrices associating software components and functions to overall system requirements. (CDRL A013)

3.3.1.2.10 The Contractor shall assist by developing and updating software documentation at a level of detail that is sufficient enough to provide a developer with comparable levels of experience and the ability to further develop software products IAW *American Society for Testing and Materials (ASTM) manual A*. The software documentation shall include, at a minimum, the following: a complete description of all software-developed instructions for using routines and programs supplied and delivered; a listing of each program supplied and delivered; source code; and all command line, keys, passphrases, configurations, and executables required for the Government's continued development and use.

3.3.1.3 Software Testing Support.

3.3.1.3.1 The Contractor shall support the development of software test cases, procedures, and test case results to include, but not be limited to, regression training, suitability testing, and acceptance testing. (CDRL A012)

3.3.1.3.2 The Contractor shall support the preparation of test reports. (CDRL A019)

3.3.1.3.3 The Contractor shall support the design, development, and fabrication of prototype testing software, articles, and equipment.

3.3.1.3.4 The Contractor shall support the process of analyzing software integration on new and existing software code.

3.3.2 IT Support.

3.3.2(a) (PROC) Provide IT support to modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks during the production phase of a project's life cycle. Applicable to paragraphs 3.3.2.1 – 3.3.2.4.

3.3.2(b) (O&M) Provide IT support to operate, maintain, refurbish, overhaul, train, and perform in-service engineering on software, computers, and networks. Applicable to paragraphs 3.3.2.1 – 3.3.2.4.

3.3.2(c) (FMS) Provide IT support to develop, prototype, evaluate, establish performance capabilities, operate, maintain, refurbish, overhaul, train, perform in-service engineering, modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks for FMS customers. Applicable to paragraphs 3.3.2.1 – 3.3.2.4.

3.3.2(d) (RDT&E) Provide IT support to research, develop, prototype, evaluate, establish performance capabilities, support associated testing and evaluation efforts, and support training in order to meet developmental milestones prior to operational use for software, computers, and networks during the RDT&E phase of a project's life cycle. Applicable to paragraphs 3.3.2.1 – 3.3.2.4.

3.3.2(e) (WCF) Provide IT support to develop, prototype, evaluate, establish performance capabilities, operate, maintain, refurbish, overhaul, train, perform in-service engineering, modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks during all phases of a project's life cycle. Applicable to paragraphs 3.3.2.1 – 3.3.2.4.

3.3.2.1 IT Design and Development Support.

3.3.2.1.1 The Contractor shall support the development of information system architectures for systems and systems of systems. This includes, but is not limited to, system requirements, functional diagrams, physical block diagrams, connectivity diagrams, hardware and software lists, and other architecture elements. (CDRLs A017, A018, A020, A021, A022, and A023)

3.3.2.1.2 The Contractor shall support these products through standard architecture tools and/or through Model-Based System Engineering (MBSE) tools.

3.3.2.1.3 The Contractor shall support development of information system device hardware, software, and firmware configurations.

3.3.2.1.4 The Contractor shall support research and development of tools, procedures, applications, and scripts to integrate baseline information technologies configurations into multiple similar or diverse platforms.

3.3.2.1.5 The Contractor shall support analyzing life cycle and maintenance operations through development of networking tools and procedures for systems equipment.

3.3.2.2 IT Testing and Integration Support.

3.3.2.2.1 The Contractor shall support integration and upgrades of mission system components to include, but not be limited to, radios, antennas, servers, switches, routers, virtual machines, cryptographic items, Wide Area Network (WAN) components, LAN components, Cross Domain Systems (CDS), access points, wireless controllers, and related support equipment and peripherals into a cohesive system. (CDRLs A017 and A024)

3.3.2.2.2 The Contractor shall support researching, analyzing, and developing system interfaces to connect systems to external systems to include assessing areas such as satellite connectivity, WANs, telephony systems, data repositories, secure enclaves, and other distributed DoD and non-DoD systems. (CDRL A017)

3.3.2.3 IT Administration and Maintenance Support.

3.3.2.3.1 The Contractor shall support IT administration duties.

3.3.2.3.2 The Contractor shall support the installation, configuration, and troubleshooting co of routers, switches, firewalls, servers, desktop computers, laptop computers, peripheral equipment, and other IT devices.

3.3.2.3.3 The Contractor shall support analyzing and updating repositories of system attributes to include, but not be limited to, networking scripts, configuration files, credentials, address mapping, access control lists, PKI, policies, procedures, and archives.

3.3.2.3.4 The Contractor shall support researching, analyzing, and administering Software Integration Laboratory (SIL) networks to include administering software updates patches and security updates.

3.3.2.3.5 The Contractor shall support configuration analysis and verification of SIL networks.

3.3.2.4 Network Documentation Support.

3.3.2.4.1 The Contractor shall support the development of IT architecture documentation. (CDRL A020)

3.3.2.4.2 The Contractor shall support the development of system designs and drawings for IT architecture. (CDRL A020)

3.3.2.4.3 The Contractor shall support the Government with the development of SETR documentation as it relates to the development of network solutions.

3.3.2.4.4 The Contractor shall support researching, analyzing, and developing configuration control for systems to include documenting system parameters, architectures, interfaces, and scripting languages.

3.3.2.4.5 The Contractor shall support the development of verification matrices, associating network components and functions to overall system requirements. (CDRL A013)

3.3.2.4.6 The Contractor shall support reviewing and analyzing product drawings and associated lists to verify technical adequacy and conformance to technical requirements. (CDRLs A023 and A025)

3.3.2.4.7 The Contractor shall support analyzing new system concepts prior to incorporation into Technical Data Packages (TDPs), which are established baselines of the networks. (CDRL A024)

3.3.3 Cyber Assessment, Accreditation, and Resilience Support.

3.3.3(a) (PROC) Provide cyber assessment, accreditation, and resilience support to modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks during the production phase of a project's life cycle. Applicable to paragraphs 3.3.3.1 – 3.3.3.2.

3.3.3(b) (O&M) Provide cyber assessment, accreditation, and resilience support to operate, maintain, refurbish, overhaul, train, and perform in-service engineering on software, computers, and networks. Applicable to paragraphs 3.3.3.1 – 3.3.3.2.

3.3.3(c) (FMS) Provide cyber assessment, accreditation, and resilience support to develop, prototype, evaluate, establish performance capabilities, operate, maintain, refurbish, overhaul, train, perform in-service engineering, modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks for FMS customers. Applicable to paragraphs 3.3.3.1 – 3.3.3.2.

3.3.3(d) (RDT&E) Provide cyber assessment, accreditation, and resilience support to research, develop, prototype, evaluate, establish performance capabilities, support associated testing and evaluation efforts, and support training in order to meet developmental milestones prior to operational use for software, computers, and networks during the RDT&E phase of a project's life cycle. Applicable to paragraphs 3.3.3.1 – 3.3.3.2.

3.3.3(e) (WCF) Provide cyber assessment, accreditation, and resilience support to develop, prototype, evaluate, establish performance capabilities, operate, maintain, refurbish, overhaul, train, perform in-service engineering, modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks during all phases of a project's life cycle. Applicable to paragraphs 3.3.3.1 – 3.3.3.2.

3.3.3.1 Cyber Assessment and Accreditation Support.

3.3.3.1.1 The Contractor shall support the development of a system Assessment and Authorization (A&A) plan to include, but not be limited to, Authority to Operate (ATO), Interim ATO (IATO), Interim Authority to Test (IATT), Interim Authority to Connect (IATC), Authority to Connect (ATC), and Authority to Develop (ATD). (CDRL A026)

3.3.3.1.2 The Contractor shall support the development of documentation required for A&A submission. This includes, but is not limited to, system security plans, security controls analysis report, cyber risk assessment reports, cybersecurity scans, STIG checklists, boundary diagrams, Plan of Action and Milestones (POA&M) inputs, cyber mitigations, and cyber-related policies and procedures. (CDRL A026)

3.3.3.1.3 The Contractor shall support Interim Progress Reviews on all A&A documentation at intervals of 190, 105, and 85 calendar days prior to reaccreditation. (CDRL A026)

3.3.3.1.4 The Contractor shall support the development of Cybersecurity Implementation Plans for systems with qualifying classification levels. (CDRL A027)

3.3.3.1.5 The Contractor shall support the analysis of software, network, and system cybersecurity to include researching existing systems and components to verify their cybersecurity posture.

3.3.3.2 Cyber Resilience and Protection Support.

3.3.3.2.1 The Contractor shall support the identification and mitigation IAW *Intelligence Community Directive (ICD) 503, Intelligence Community Information Environment Risk Management* of vulnerabilities for aircraft and weapon Real Time Operating Systems (RTOSs) in a system of systems warfare environment with often intermittent or indirect connectivity to other systems. (CDRL A028)

3.3.3.2.2 The Contractor shall support the identification and mitigation of vulnerabilities for networks, simulators, maintenance laptops, mission loaders, critical physical and industrial control system interfaces with air vehicles to include, but not be limited to, aircraft launch and recovery equipment, power and navigation umbilicals, and supply chain management. (CDRL A028)

3.3.3.2.3 The Contractor shall support the research, identification, and mitigation of cyber vulnerabilities to increase cyber resiliency of systems. (CDRL A028)

3.3.3.2.4 The Contractor shall support dynamic reconfiguration, compilation, and re-hosting for RTOS, embedded Platform Information Technologies (PITs), supervisory and data acquisition control systems, operational warfighting networks, and maintenance and logistics reach back networks.

3.3.3.2.5 The Contractor shall support size, weight, and power sensitive cyber resiliency solutions for MLS and RTOSs.

3.3.3.2.6 The Contractor shall support detection, protection, incident response, recovery, and infrastructure. Recovery could include, but not be limited to, malware or other effects on aircraft and weapon system RTOSs. Infrastructure could include tools, response kits, and methods.

3.3.3.2.7 The Contractor shall support researching, analyzing, identifying, and implementing applicable STIGs for system components and systems.

3.3.3.2.8 The Contractor shall support testing of STIG implementations for use in system verification and accreditation activities.

3.3.3.2.9 The Contractor shall support researching, evaluating, developing, and implementing system controls to reduce vulnerabilities and the mitigation of discrepancy reports.

3.3.3.2.10 The Contractor shall support assessing and identifying vulnerabilities and developing mitigation strategies for systems. (CDRL A028)

3.3.3.2.11 The Contractor shall support identifying, resolving, developing, and integrating software solutions to security vulnerabilities of existing Government-specified software code.

3.3.3.2.12 The Contractor shall support documenting and reporting vulnerabilities using the appropriate reporting system as stated in the program's Security Classification Guide (SCG), as applicable.

3.3.3.2.13 The Contractor shall support cyber-related continuous monitoring of specified systems through various stages of the system acquisition life cycle.

3.3.4 Product Assurance, Training, Documentation, and Configuration Management (CM) Support.

3.3.4(a) (PROC) Provide product assurance, training, documentation, and CM support to modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks during the production phase of a project's life cycle. Applicable to paragraphs 3.3.4.1 – 3.3.4.3.

3.3.4(b) (O&M) Provide product assurance, training, documentation, and CM support to operate, maintain, refurbish, overhaul, train, and perform in-service engineering on software, computers, and networks. Applicable to paragraphs 3.3.4.1 – 3.3.4.3.

3.3.4(c) (FMS) Provide product assurance, training, documentation, and CM support to develop, prototype, evaluate, establish performance capabilities, operate, maintain, refurbish, overhaul, train, perform in-service engineering, modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks for FMS customers. Applicable to paragraphs 3.3.4.1 – 3.3.4.3.

3.3.4(d) (RDT&E) Provide product assurance, training, documentation, and CM support to research, develop, prototype, evaluate, establish performance capabilities, support associated testing and evaluation efforts, and support training in order to meet developmental milestones prior to operational use for software, computers, and networks during the RDT&E phase of a project's life cycle. Applicable to paragraphs 3.3.4.1 – 3.3.4.3.

3.3.4(e) (WCF) Provide product assurance, training, documentation, and CM support to develop, prototype, evaluate, establish performance capabilities, operate, maintain, refurbish, overhaul, train, perform in-service engineering, modify, modernize, retrofit, integrate, and conduct initial training and production engineering on software, computers, and networks during all phases of a project's life cycle. Applicable to paragraphs 3.3.4.1 – 3.3.4.3.

3.3.4.1 Product Assurance Support.

3.3.4.1.1 The Contractor shall support monitoring the software and information systems development processes and methods to ensure quality.

3.3.4.1.2 The Contractor shall support monitoring software and information systems for proper operation, use, maintenance, and disposal IAW DON security policy and practices.

3.3.4.1.3 The Contractor shall support tracking and controlling changes in software, information systems, and supporting data to ensure version control; establish baselines; implement a change process in order to provide configuration status accounting, configuration auditing, build management, process management, environment management, and defect tracking.

3.3.4.1.4 The Contractor shall support the upkeep, configuration, and reliable operation of software, computers, and networks.

3.3.4.1.5 The Contractor shall support planning for and responding to service outages.

3.3.4.1.6 The Contractor shall support assurance of software IAW *NAVAIR SWP4P00-047.3, Cybersafe Certification*.

3.3.4.1.7 The Contractor shall support review and analysis of emerging information systems technologies, capabilities, and requirements to maintain systems applicability.

3.3.4.2 Training, Documentation, and CM Support.

3.3.4.2.1 The Contractor shall support the development and delivery of training and documentation related to regulations, policy directives, Standard Operating Procedures (SOPs), user guides, and other content areas related to software, computers, and networks. (CDRL A029)

3.3.4.2.2 The Contractor shall support the preparation and delivery of technical reports and memoranda outlining technical research completed, potential investment strategies, and technology development plans. (CDRL A030)

3.3.4.2.3 The Contractor shall support the development and delivery of training products and manuals for the O&M of software and information systems developed. (CDRLs A031 and A032)

3.3.4.2.4 The Contractor shall support the preparation and delivery of exhibits and displays associated with software, computer, and networks developed to support avionics and weapons systems. (CDRL A033)

3.3.4.2.5 The Contractor shall support researching, analyzing, identifying, and implementing mitigation methods to reduce system discrepancies, non-compliance attributes, and “bugs” in the system baseline.

3.3.4.2.6 The Contractor shall support assessing, identifying, troubleshooting, and repairing equipment failures in both laboratory and field environments. (CDRL A034)

3.3.4.2.7 The Contractor shall support CM to include, but not be limited to, Physical Configuration Audits (PCAs), Functional Configuration Audits (FCAs), and Configuration Item

Verification Reviews (CIVRs) by participating in the processes, verifying results, and developing recommendations for future capabilities. (CDRL A030)

3.3.4.3 Software and Information Systems Maintenance Support.

3.3.4.3.1 The Contractor shall support troubleshooting of fielded software and information system products.

3.3.4.3.2 The Contractor shall support updates to technical documentation, operation manuals, software documentation, and user guides as a result of software and information systems maintenance updates.

3.3.4.3.3 The Contractor shall support maintaining and conducting routine maintenance and upgrade activities.

3.3.4.3.4 The Contractor shall support on-site technical support. (CDRL A034)

3.3.4.3.5 The Contractor shall support maintaining a database of trouble tickets to include specifics to issues observed, ticket statuses, and steps executed to resolve each issue. (CDRL A034)

3.4 Personnel Qualifications.

3.4.1 The following defines the minimum education and experience for each non-Service Contract Act (non-SCA) labor category and further describes the function of each labor category. The Contractor shall be responsible for employing personnel having at least the minimum level of education, training, and experience (including specialized experience) as stated under each labor category specified herein. All personnel identified in paragraph. 3.2.3.1 must have the minimum security clearance indicated.

3.4.2 Key personnel are those who will be performing in the key labor categories as specified for applicable labor categories below. All other labor categories and Full Time Equivalent (FTE) employees (defined as 1,920 hours) are considered non-key. Key personnel are subject to the substitution restrictions within CTXT.237-9501, Additional or Substitution of Key Personnel (Services). Authorized Key Personnel are identified within Section J, Attachment 02 – List of Authorized Key Personnel.

KEY LABOR CATEGORY	LEVEL	BUREAU OF LABOR STATISTICS (BLS) STANDARD OCCUPATIONAL CLASSIFICATION (SOC) CODE	NUMBER OF KEY PERSONNEL
Engineers, All Other (Systems Engineer)	Senior	17-2199	One
General and Operations Managers (Program Manager)	Senior	11-1021	One

Software Developers	Advanced	15-1252, Cybersecurity Work Force (CSWF) Role 621	One

3.4.2.1 Definitions.

3.4.2.1.1 “Years of experience” shall mean full, productive years of experience.

3.4.2.1.2 “Productive years” shall mean 52 weeks of work reduced by reasonable amounts of time for holiday, annual, and sick leave.

3.4.2.1.3 “Part-time” shall mean if participation was part-time or if less than one-half of the standard work week was spent performing qualifying functions. The actual time spent performing qualifying functions may be accumulated to arrive at full years of experience.

3.4.2.1.4 “College degree” shall mean all degrees shall be obtained from an accredited college or university as recognized by the U.S. Department of Education (ED). This includes Associate of Arts (A.A.), Associate of Science (A.S.), Bachelor of Arts (B.A.), Bachelor of Science (B.S.), Master of Arts (M.A.), Master of Science (M.S.), and/or Doctorate (Ph.D.) degrees.

3.4.2.1.5 “Degree major” shall mean the specific major field required, which will be noted under the applicable labor category.

3.4.2.1.6 “Engineering or engineering discipline,” when used in relation to educational or work experience requirements, shall mean any of the following specific subjects, disciplines, or areas of work experience only: aerospace, computer, electrical, electronics, mechanical, information systems, engineering technology, network, physics, software, systems engineering, or industrial engineering.

3.4.2.1.7 “Technical discipline,” when used in relation to educational or work experience requirements, shall mean any of the following specific subjects, disciplines, or areas of work experience only: aerospace, electrical, electronics, mathematics, mechanical, physics, information systems, or engineering technology.

3.4.2.1.8 “Business or business discipline,” when used in relation to educational or work experience requirements, shall mean any of the following specific subjects, disciplines, or areas of work experience only: business administration, business management, project management, economics, finance, international business, legal studies, political science, management studies, English, marketing, or accounting.

3.4.2.1.9 “Computer science or IT discipline,” when used in relation to educational or work experience requirements, shall mean any of the following specific subjects, disciplines, or areas of work experience only: information or computer science, computer engineering, software engineering, network engineering, information systems, cybersecurity, or management information systems technology.

3.4.2.1.10 “Technical certification training” shall mean the specified certification training noted under the applicable labor category.

3.4.2.1.11 “Active or current certification” shall mean individuals must be able to demonstrate that they possess a valid certification, per vendor recertification standards, at the time of proposal submission and at the time of TO award.

3.4.3 Non-SCA and non-CSWF labor categories experience and education level definitions. If required, specialized experience and associated years of that particular experience are specified under the applicable labor category qualifications.

3.4.3.1 Junior. A junior level person within a labor category has less than three years of experience and a bachelor’s degree or a qualifying substitution, if identified. However, experience may exceed three years if performing a junior level function. A junior level person is responsible for assisting more senior positions and/or performing functional duties under the oversight of more senior positions.

3.4.3.2 Journeyman. A journeyman level person within a labor category has three or more years of experience and a bachelor’s degree or a qualifying substitution, if identified. A journeyman level person typically performs all functional duties independently.

3.4.3.3 Senior. A senior level person has 10 or more years of experience and a master’s degree, or a qualifying substitution, if identified. A senior level person typically works on high-visibility or mission-critical aspects of a given program and performs all functional duties independently. A senior level person may oversee the efforts of less senior staff and/or be responsible for the efforts of all staff assigned to a specific job.

3.4.3.4 Qualification Substitution Chart. Except for the labor categories noted in SOW paragraph 3.4.3.5, the following chart provides allowable qualification substitutions.

Bachelor’s Degree	Associate’s degree plus four years of relevant experience exceeding the experience requirement stated in SOW paragraph 3.4.3 for the applicable labor category.	OR	High School diploma or General Educational Development (GED) diploma plus six years of relevant work experience exceeding the experience requirement stated in SOW paragraph 3.4.3 for the applicable labor category.
Master’s Degree	Bachelor’s degree plus four years of relevant experience exceeding the experience requirement stated in SOW paragraph 3.4.3 for the applicable labor category.	OR	Associate’s degree plus six years of relevant experience exceeding the experience requirement stated in SOW paragraph 3.4.3 for the applicable labor category.

3.4.3.5 Exceptions to Qualification Substitution Chart. The labor categories stated below require a minimum of a bachelor’s degree in the discipline stated for the labor category. If a master's degree is required for any of the labor categories listed below, a substitution of a bachelor's degree plus four years of relevant experience exceeding the experience requirement stated in paragraph 3.4.3 for the applicable labor category is allowable.

a) Aerospace Engineers, Journeyman.

b) Aerospace Engineers, Senior.

c) Electrical Engineers, Junior.

d) Electrical Engineers, Journeyman.

e) Electrical Engineers, Senior.

f) Electronics Engineers, Except Computer, Junior.

g) Electronics Engineers, Except Computer, Journeyman.

h) Electronics Engineers, Except Computer, Senior.

i) Engineers, All Other (Systems Engineer), Junior.

j) Engineers, All Other (Systems Engineer), Journeyman.

k) Engineers, All Other (Systems Engineer), Senior.

l) General and Operations Managers (Program Manager), Senior.

m) Mechanical Engineers, Junior.

n) Mechanical Engineers, Journeyman.

o) Mechanical Engineers, Senior.

3.4.3.6 Labor Qualifications. The following lists the minimum labor category education and experience requirements, BLS SOC Code, and functional descriptions for each labor category.

3.4.3.6.1 Aerospace Engineers, Journeyman, 17-2011.

Function: Perform engineering duties in designing, constructing, and testing aircraft, and related mission systems. May conduct basic and applied research to evaluate adaptability of materials and equipment to aircraft design and manufacture. May recommend improvements in testing equipment and techniques.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Bachelor’s degree in an Engineering discipline.

3.4.3.6.2 Aerospace Engineers, Senior, 17-2011.

Function: Perform engineering duties in designing, constructing, and testing aircraft, and related mission systems. May conduct basic and applied research to evaluate adaptability of materials and equipment to aircraft design and manufacture. May recommend improvements in testing equipment and techniques.

Required Experience: Of the minimum 10 years of experience required, at least 5 of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Master's degree in an Engineering discipline or qualifying substitution.

3.4.3.6.3 Avionics Technicians, Junior, 49-2091.

Function: Install, inspect, test, adjust, or repair the electronic systems in aircraft, including communication, navigation, radar, and flight control systems, ensuring safe and reliable operation in aircraft and related mission systems.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor's degree in an Engineering, Technical, or Computer Science/IT discipline.

3.4.3.6.4 Avionics Technicians, Journeyman, 49-2091.

Function: Install, inspect, test, adjust, or repair the electronic systems in aircraft, including communication, navigation, radar, and flight control systems, ensuring safe and reliable operation in aircraft and related mission systems.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to avionics systems within the DoD.

Required Education: Bachelor's degree in an Engineering, Technical, or Computer Science/IT discipline or qualifying substitution.

3.4.3.6.5 Avionics Technicians, Senior, 49-2091.

Function: Install, inspect, test, adjust, or repair the electronic systems in aircraft, including communication, navigation, radar, and flight control systems, ensuring safe and reliable operation in aircraft and related mission systems.

Required Experience: Of the minimum 10 years of experience required, at least 5 of those years must be specific to avionics systems within the DoD.

Required Education: Master's degree in an Engineering, Technical, or Computer Science/IT discipline or qualifying substitution.

3.4.3.6.6 Drafters, All Other (MBSE), Junior, 17-3019.

Function: Research, design, develop, and perform drafting functions utilizing MBSE methodologies, tools, templates, and processes to support MBSE efforts by creating and maintaining digital models of systems to simulate, test, and analyze the behavior of the system, aiding in design decisions and requirements verification. Recommend adjustments for MBSE methodologies, tools, templates, and processes to improve efficiencies in project execution.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor's degree in an Engineering, Technical, or Computer Science/IT discipline or qualifying substitution.

3.4.3.6.7 Drafters, All Other (MBSE), Journeyman, 17-3019.

Function: Research, design, develop, and perform drafting functions utilizing MBSE methodologies, tools, templates, and processes to support MBSE efforts by creating and maintaining digital models of systems to simulate, test, and analyze the behavior of the system, aiding in design decisions and requirements verification. Recommend adjustments for MBSE methodologies, tools, templates, and processes to improve efficiencies in project execution.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Bachelor's degree in an Engineering, Technical, or Computer Science/IT discipline or qualifying substitution.

3.4.3.6.8 Drafters, All Other (MBSE), Senior, 17-3019.

Function: Research, design, develop, and perform drafting functions utilizing MBSE methodologies, tools, templates, and processes to support MBSE efforts by creating and maintaining digital models of systems to simulate, test, and analyze the behavior of the system, aiding in design decisions and requirements verification. Recommend adjustments for MBSE methodologies, tools, templates, and processes to improve efficiencies in project execution.

Required Experience: Of the minimum 10 years of experience required, at least 5 of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Master's degree in an Engineering, Technical, or Computer Science/IT discipline or qualifying substitution.

3.4.3.6.9 Electrical Engineers, Junior, 17-2071.

Function: Research, design, develop, test, or supervise the manufacturing and installation of electrical equipment, components, or systems for commercial, industrial, military, or scientific use.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor’s degree in an Engineering discipline.

3.4.3.6.10 Electrical Engineers, Journeyman, 17-2071.

Function: Research, design, develop, test, or supervise the manufacturing and installation of electrical equipment, components, or systems for commercial, industrial, military, or scientific use.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Bachelor’s degree in an Engineering discipline.

3.4.3.6.11 Electrical Engineers, Senior, 17-2071.

Function: Research, design, develop, test, or supervise the manufacturing and installation of electrical equipment, components, or systems for commercial, industrial, military, or scientific use.

Required Experience: Of the minimum 10 years of experience required, at least 5 of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Master’s degree in an Engineering discipline or qualifying substitution.

3.4.3.6.12 Electronics Engineers, Except Computer, Junior, 17-2072.

Function: Research, design, develop, or test electronic components and systems for commercial, industrial, military, or scientific use employing knowledge of electronic theory and material properties. Design electronic circuits and components for use in fields such as telecommunications, aerospace guidance and propulsion control, acoustics, or instruments and controls.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor’s degree in an Engineering discipline.

3.4.3.6.13 Electronics Engineers, Except Computer, Journeyman, 17-2072.

Function: Research, design, develop, or test electronic components and systems for commercial, industrial, military, or scientific use employing knowledge of electronic theory and materials properties. Design electronic circuits and components for use in fields such as telecommunications, aerospace guidance and propulsion control, acoustics, or instruments and controls.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Bachelor's degree in an Engineering discipline.

3.4.3.6.14 Electronics Engineers, Except Computer, Senior, 17-2072.

Function: Research, design, develop, or test electronic components and systems for commercial, industrial, military, or scientific use employing knowledge of electronic theory and materials properties. Design electronic circuits and components for use in fields such as telecommunications, aerospace guidance and propulsion control, acoustics, or instruments and controls.

Required Experience: Of the minimum 10 years of experience required, at least 5 of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Master's degree in an Engineering discipline or qualifying substitution.

3.4.3.6.15 Engineers, All Other (Systems Engineer), Junior, 17-2199.

Function: Provide systems engineering direction and guidance for design, development, integration, and interface design analysis, installation, integration, fielding, field analysis, operation, maintenance, and testing of information processing and other systems.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor's degree in an Engineering discipline.

3.4.3.6.16 Engineers, All Other (Systems Engineer), Journeyman, 17-2199.

Function: Provide systems engineering direction and guidance for design, development, integration, and interface design analysis, installation, integration, fielding, field analysis, operation, maintenance, and testing of information processing and other systems.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Bachelor's degree in an Engineering discipline.

3.4.3.6.17 Engineers, All Other (Systems Engineer), Senior, 17-2199, KEY (One).

Function: Provide systems engineering direction and guidance for design, development, integration, and interface design analysis, installation, integration, fielding, field analysis, operation, maintenance, and testing of information processing and other systems.

Required Experience: Of the minimum 10 years of experience required, at least 5 of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Master’s degree in an Engineering discipline or qualifying substitution.

3.4.3.6.18 Financial Specialists, All Other, Junior, 13-2099.

Function: Prepare and provide cost estimates for Government projects. Conduct analyses of information involving Government projects and financial data of private institutions. Prepare and provide recurring deliverables outlining contractual financial progress.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor’s degree in a Business discipline or qualifying substitution.

3.4.3.6.19 Financial Specialists, All Other, Journeyman, 13-2099.

Function: Prepare and provide cost estimates for Government projects. Conduct analyses of information involving Government projects and financial data of private institutions. Prepare and provide recurring deliverables outlining contractual financial progress.

Required Experience: Of the minimum three years of experience required, at least two of those years must be related to DON financial management and utilizing contractor invoicing systems.

Required Education: Bachelor’s degree in a Business discipline or qualifying substitution.

3.4.3.6.20 General and Operations Managers (Program Manager), Senior, 11-1021, KEY (One).

Function: Plan, direct, or coordinate the operations of public or private sector organizations, overseeing multiple departments or locations. Duties and responsibilities include formulating policies, managing daily operations, and planning the use of materials and human resources, but are too diverse and general in nature to be classified in any one functional area of management or administration such as personnel, purchasing, or administrative services. Usually manage through subordinate supervisors. Act as the overall lead, manager, and administrator for the entire contracted effort. Serve as the primary interface and point of contact with the Government COR on technical and project issues. Oversee contractor execution of the TO requirements. Manage acquisition and employment of project resources.

Required Experience: Of the minimum 10 years of experience required, at least 7 of those years must include supervising and providing recommendations to all areas of program management, systems engineering, major system acquisitions, and financial management.

Required Education: Master’s degree in an Engineering, Technical, Business, or Computer Science/IT discipline or qualifying substitution.

3.4.3.6.21 Logisticians, Junior, 13-1081.

Function: Analyze and coordinate the ongoing logistical functions of a firm or organization. Responsible for the entire life cycle of a product, including acquisition, distribution, internal allocation, delivery, and final disposal of resources.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor's degree in Engineering, Technical, or Business degree or qualifying substitution.

3.4.3.6.22 Logisticians, Journeyman, 13-1081.

Function: Analyze and coordinate the ongoing logistical functions of a firm or organization. Responsible for the entire life cycle of a product, including acquisition, distribution, internal allocation, delivery, and final disposal of resources.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Bachelor's degree in an Engineering, Technical, or Business discipline or qualifying substitution.

3.4.3.6.23 Logisticians, Senior, 13-1081.

Function: Analyze and coordinate the ongoing logistical functions of a firm or organization. Responsible for the entire life cycle of a product, including acquisition, distribution, internal allocation, delivery, and final disposal of resources.

Required Experience: Of the minimum 10 years of experience required, at least 3 of those years must be related to DoD software, computer, and/or network systems.

Required Education: Master's degree in an Engineering, Technical, or Business discipline or a qualifying substitution.

3.4.3.6.24 Mechanical Engineers, Junior, 17-2141.

Function: Perform engineering duties in planning and designing tools, engines, machines, and other mechanically functioning equipment. Oversee installation, operation, maintenance, and repair of equipment such as centralized heat, gas, water, and steam systems.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor's degree in an Engineering discipline.

3.4.3.6.25 Mechanical Engineers, Journeyman, 17-2141.

Function: Perform engineering duties in planning and designing tools, engines, machines, and other mechanically functioning equipment. Oversee installation, operation, maintenance, and repair of equipment such as centralized heat, gas, water, and steam systems.

Required Experience: Of the minimum three years of required experience, at least two of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Bachelor’s degree in an Engineering discipline.

3.4.3.6.26 Mechanical Engineers, Senior, 17-2141.

Function: Perform engineering duties in planning and designing tools, engines, machines, and other mechanically functioning equipment. Oversee installation, operation, maintenance, and repair of equipment such as centralized heat, gas, water, and steam systems.

Required Experience: Of the minimum 10 years of required experience, at least 5 of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Master’s degree in an Engineering discipline or qualifying substitution.

3.4.3.6.27 Operations Research Analysts, Senior, 15-2031.

Function: Formulate and apply mathematical modeling and other optimizing methods to develop and interpret information that assists management with decision making, policy formulation, or other managerial functions such as developing and managing budgets and analyzing financial data based on operational insight.

Required Experience: Of the minimum 10 years of required experience, at least 5 of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Master’s degree in an Engineering, Technical, Business, or Computer Science/IT discipline or qualifying substitution.

3.4.3.6.28 Project Management Specialists, Junior, 13-1082.

Function: Analyze and coordinate the schedule, timeline, procurement, staffing, and budget of a product or service on a per-project basis. Lead and guide the work of technical staff. May serve as a point of contact for the client or customer. Perform cost/benefit analyses, quality control, and project tracking against critical path events.

Required Experience: No specific field experience required other than what is identified in the function description above.

Required Education: Bachelor’s degree in an Engineering, Technical, Business, or Computer Science/IT discipline or qualifying substitution.

3.4.3.6.29 Project Management Specialists, Journeyman, 13-1082.

Function: Analyze and coordinate the schedule, timeline, procurement, staffing, and budget of a product or service on a per-project basis. Lead and guide the work of technical staff. May serve as a point of contact for the client or customer. Perform cost/benefit analyses, quality control, and project tracking against critical path events.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Bachelor's degree in an Engineering, Technical, or Business discipline or qualifying substitution.

3.4.3.6.30 Project Management Specialists, Senior, 13-1082.

Function: Analyze and coordinate the schedule, timeline, procurement, staffing, and budget of a product or service on a per-project basis. Lead and guide the work of technical staff. May serve as a point of contact for the client or customer. Perform cost/benefit analyses, quality control, and project tracking against critical path events.

Required Experience: Of the minimum 10 years of experience required, at least 6 of those years must be specific to software, computer, and/or network systems within the DoD.

Required Education: Master's degree in an Engineering, Technical, or Business discipline or qualifying substitution.

3.4.4 Cybersecurity Workforce (CSWF) Designation Positions. The labor category levels for designated positions are specified as Entry/Basic, Intermediate, and Advanced IAW the following definitions and CSWF qualifications under the Navy Credentialing Opportunities On-Line's CWF Workforce Model, which can be found at https://www.cool.osd.mil/usn/cswf/index.html.

3.4.5 IAW *SECNAVINST 5239.3B, Department of the Navy Information Assurance Policy SECNAVINST 5239.25, Department of the Navy Cyberspace Workforce Qualification and Management Program*, procedures surrounding the qualification of the Department of the Navy (DON) Cyberspace IT/CSWF are discussed in *DoD Directive 8140.01, Cyberspace Workforce Management* (or any subsequent revisions), which replaced *DoD Directive 8570.01-M* and *NIST Special Publication 800-181 and* IAW *DoDI 8140.02, Identification, Tracking, and Reporting of Cyberspace Workforce Requirements* and *DOD Manual 8140.03 Cyberspace Workforce Qualification and Management Program, 15 Feb 2023.*

3.4.5.1 Work Role Qualification Matrix

The matrix below documents the standard experience and education level definitions for CSWF labor categories. The matrix demonstrates that education, or training, or certification requirements can fulfill the CSWF labor category foundational qualifications. The foundational qualifications are the equivalent of what would normally be considered a required education. Additional specific required experience and certifications (if any) for each CSWF labor category is stated below in paragraph 3.4.6 and its subparagraphs.

BASIC	INTERMEDIATE	ADVANCED
Meets the identified CSWF Education qualification stated within Attachment 04 - CSWF Foundational Qualification Options.	Meets the identified CSWF Education qualification stated within Attachment 04 - CSWF Foundational Qualification Options.	Meets the identified CSWF Education qualification stated within Attachment 04 - CSWF Foundational Qualification Options.
OR	**OR**	**OR**
Meets the identified CSWF Training qualification stated within Attachment 04 - CSWF Foundational Qualification Options.	Meets the identified CSWF Training qualification stated within Attachment 04 - CSWF Foundational Qualification Options.	Meets the identified CSWF Training qualification stated within Attachment 04 - CSWF Foundational Qualification Options.
OR	**OR**	**OR**
Meets the identified CSWF Certification qualification stated within Attachment 04 - CSWF Foundational Qualification Options.	Meets the identified CSWF Certification qualification stated within Attachment 04 - CSWF Foundational Qualification Options.	Meets the identified CSWF Certification qualification stated within Attachment 04 - CSWF Foundational Qualification Options.
AND	**AND**	**AND**
Has less than three years of experience performing work related to the labor category functional description OR meets the identified military equivalent On-the-Job (OTJ) Training requirement.	Has three or more years of experience performing work related to the labor category functional description OR meets the identified military equivalent OTJ Training requirement.	Has at least seven years of experience performing work related to the labor category functional description OR meets the identified military equivalent OTJ requirement.

3.4.6 CSWF Services Labor Qualifications.

Attachment 04 - CSWF Foundational Qualification Options includes education, training and certification options that are customized for this TO for each CSWF labor category. Contractors shall adhere to the options in this attachment for each CSWF labor category. Any certifications listed in the SOW paragraphs below are required and are not part of the CSWF Foundational Qualification Options. The following lists the minimum labor category education and experience requirements, BLS SOC Code, telework work eligibility, and functional descriptions for each labor category.

3.4.6.1 Computer and Information Research Scientists, Intermediate, 15-1221.

Function: Conduct research into fundamental computer and information science as theorists, designers, or inventors. Develop solutions to problems in the field of computer hardware and software.

Work Role Identification: 661

Telework Work Eligibility: Full-time and Hybrid Telework are dependent on specific tasking and will be authorized by the COR.

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to DoD systems.

3.4.6.2 Computer and Information Research Scientists, Advanced, 15-1221.

Function: Conduct research into fundamental computer and information science as theorists, designers, or inventors. Develop solutions to problems in the field of computer hardware and software.

Work Role Identification: 661

Required Experience: Of the minimum seven years of experience required, at least two of those years must be specific to DoD systems.

3.4.6.3 Computer Hardware Engineers, Intermediate, 17-2061.

Function: Research, design, develop, or test computer or computer-related equipment for commercial, industrial, military, or scientific use. May supervise the manufacturing and installation of computer or computer-related equipment and components.

Work Role Identification: 632

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to DoD systems.

3.4.6.4 Computer Hardware Engineers, Advanced, 17-2061.

Function: Research, design, develop, or test computer or computer-related equipment for commercial, industrial, military, or scientific use. May supervise the manufacturing and installation of computer or computer-related equipment.

Work Role Identification: 632

Required Experience: Of the minimum seven years of experience required, at least two of those years must be specific to DoD systems.

3.4.6.5 Computer Network Architects, Intermediate, 15-1241.

Function: Design and implement computer and information networks, such as LANs, WANs, intranets, extranets, and other data communications networks. Perform network modeling, analysis, and planning, including analysis of capacity needs for network infrastructures. May also design network and computer security measures. May research and recommend network and data communications hardware and software.

Work Role Identification: 632

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to DoD systems.

3.4.6.6 Computer Programmers, Intermediate, 15-1251.

Function: Create, modify, and test the code and scripts that allow computer applications to run. Work from specifications drawn up by software and web developers or other individuals. May develop and write computer programs to store, locate, and retrieve specific documents, data, and information.

Work Role Identification: 621

Required Experience: Of the minimum three years of experience required, at least three years of experience must be directly related to database development, computer-based training development, and use of .NET tools, .NET Integrated Development Environment tools, asp.net, HQL Herver 2000, and HTML editor tools and at least one of those years must be specific to DoD systems.

3.4.6.7 Computer Programmers, Advanced, 15-1251.

Function: Create, modify, and test the code and scripts that allow computer applications to run. Work from specifications drawn up by software and web developers or other individuals. May develop and write computer programs to store, locate, and retrieve specific documents, data, and information.

Work Role Identification: 621

Required Experience: Of the minimum of seven years required experience, at least three years of experience must be directly related to database development, computer-based training development, and use of .NET tools, .NET Integrated Development Environment tools, asp.net, HQL Herver 2000, and HTML editor tools and at least one of those years must be specific to DoD systems.

3.4.6.8 Computer Systems Analysts, Intermediate, 15-1211.

Function: Analyze science, engineering, business, and other data processing problems to develop and implement solutions to complex applications problems, system administration issues, or network concerns. Perform systems management and integration functions, improve existing computer systems, and review computer system capabilities, workflow, and schedule limitations. May analyze or recommend commercially available software.

Work Role Identification: 461

Required Experience: Of the minimum three years of experience required, at least one of those years must be specific to DoD systems. No specific field of experience is required other than what is identified in the function description above.

3.4.6.9 Computer Systems Analysts, Advanced, 15-1211.

Function: Analyze science, engineering, business, and other data processing problems to develop and implement solutions to complex applications problems, system administration issues, or network concerns. Perform systems management and integration functions, improve existing computer systems, and review computer system capabilities, workflow, and schedule limitations. May analyze or recommend commercially available software.

Work Role Identification: 461

Required Experience: Of the minimum seven years of experience required, at least two of those years must be specific to DoD systems. No specific field of experience is required other than what is identified in the function description above.

3.4.6.10 General and Operations Managers (Tactical Network and Computer Systems Administrator), Intermediate, 11-1021.

Function: Install, configure, and maintain weapons systems airborne and air to ground data communications network and computing systems. Perform System monitoring and verify the integrity and availability of hardware, network, and server resources and systems. Analyze systems, sensors, network, and server resource consumption and control user access. Support development, installation and upgrades to systems hardware and software baselines. Assist in network emulation/virtualization, analysis, planning, and coordination between weapons systems networks, associated test and evaluation networks, and data communications hardware and software. Support the planning and coordination of the design and implementation of weapons systems data communications networks and computing systems usually at a subsystem or sub-team level. Support program leads with providing input to the identification and planning for resourcing needs.

Work Role Identification: 632

Required Experience: Of the minimum three years of required experience, at least two of those years must be specific to DoD systems.

3.4.6.11 General and Operations Managers (Tactical Network and Computer Systems Architect), Advanced, 11-1021.

Function: Plan, direct, and coordinate the design and implementation of weapons systems airborne and air to ground data communications networks and computing systems. Perform weapons systems data network emulation/virtualization, analysis, and planning. Design of weapons systems and associated network, computer security measures, and software solutions. Analyze systems, sensors, network, and server resource consumption and control user access. Oversee development, installation, and upgrades to system software and hardware baselines. Manage daily operations of network and software developers assigning priorities and managing progress. Support identification and planning for resourcing needs. Supports development of local policies and procedures driving quality and efficiency into weapon system development and sustainment.

Assist in network emulation/virtualization, analysis, planning, and coordination between weapons systems networks, associated test and evaluation networks, and data communications hardware and software.

Work Role Identification: 632

Required Experience: Of the minimum seven years of required experience, at least two of those years must be specific to DoD systems.

3.4.6.12 Information Security Analysts, Intermediate, 15-1212.

Function: Plan, implement, upgrade, or monitor security measures for the protection of computer networks and information. Assess system vulnerabilities for security risks and propose and implement risk mitigation strategies. May ensure that appropriate security controls are in place that will safeguard digital files and vital electronic infrastructure. May respond to computer security breaches and viruses.

Work Role Identification: 461

Required Experience: Of the minimum three years of experience required, at least two of those years must be specific to DoD systems.

3.4.6.13 Information Security Analysts, Advanced, 15-1212.

Function: Plan, implement, upgrade, or monitor security measures for the protection of computer networks and information. Assess system vulnerabilities for security risks and propose and implement risk mitigation strategies. May ensure that appropriate security controls are in place that will safeguard digital files and vital electronic infrastructure. May respond to computer security breaches and viruses.

Work Role Identification: 461

Required Experience: Of the ten plus years of required experience, at least seven years of experience must be directly related to the design, analysis, test procedure generation, test execution, and implementation of security mechanisms on DoD networks.

3.4.6.14 Network and Computer Systems Administrators, Basic, 15-1244.

Function: Install, configure, and maintain an organization's LAN, WAN, data communications network, operating systems, and physical and virtual servers. Perform system monitoring and verify the integrity and availability of hardware, network, and server resources and systems. Review system and application logs and verify completion of scheduled jobs to include system backups. Analyze network and server resource consumption and control user access. Install and upgrade software and maintain software licenses. May assist in network modeling, analysis, planning, and coordination between network and data communications hardware and software.

Work Role Identification: 441

Required Experience: No specific field experience required other than what is identified in the function description above.

3.4.6.15 Network and Computer Systems Administrators, Intermediate, 15-1244.

Function: Install, configure, and maintain an organization's LAN, WAN, data communications network, operating systems, and physical and virtual servers. Perform system monitoring and verify the integrity and availability of hardware, network, and server resources and systems. Review system and application logs and verify completion of scheduled jobs to include system backups. Analyze network and server resource consumption and control user access. Install and upgrade software and maintain software licenses. May assist in network modeling, analysis, planning, and coordination between network and data communications hardware and software.

Work Role Identification: 441

Required Experience: Of the minimum three years of required experience, at least two of those years must be specific to DoD systems.

3.4.6.16 Operations Research Analysts (Cyber Analyst/Modeler), Intermediate, 15-2031.

Function: Formulate and apply mathematical modeling and other optimizing methods to develop and interpret information that assists management with decision-making, policy formulation, or other managerial functions. May collect and analyze data and develop decision support software, services, or products. May develop and supply optimal time, cost, or logistics networks for program evaluation, review, or implementation.

Work Role Identification: 612

Required Experience: Of the minimum three years of required experience, at least two of those years must be specific to DoD systems.

3.4.6.17 Operations Research Analysts (Cyber Analyst/Modeler), Advanced, 15-2031.

Function: Formulate and apply mathematical modeling and other optimizing methods to develop and interpret information that assists management with decision-making, policy formulation, or other managerial functions. May collect and analyze data and develop decision support software, services, or products. May develop and supply optimal time, cost, or logistics networks for program evaluation, review, or implementation.

Work Role Identification: 612

Required Experience: Of the minimum seven years of required experience, at least two of those years must be specific to DoD systems.

3.4.6.18 Software Developers, Intermediate, 15-1252.

Function: Research, design, and develop computer and network software or specialized utility programs. Analyze user needs and develop software solutions, applying principles and techniques of computer science, engineering, and mathematical analysis. Update software or enhance existing software capabilities. May work with computer hardware engineers to integrate hardware and software systems and to develop specifications and performance requirements. May maintain databases within an application area, working individually or coordinating database development as part of a team.

Work Role Identification: 621

Required Experience: Of the minimum three years of required experience, at least two of those years must be specific to DoD systems.

3.4.6.19 Software Developers, Advanced, 15-1252, KEY (One)

Function: Research, design, and develop computer and network software or specialized utility programs. Analyze user needs and develop software solutions, applying principles and techniques of computer science, engineering, and mathematical analysis. Update software or enhance existing software capabilities. May work with computer hardware engineers to integrate hardware and software systems and to develop specifications and performance requirements. May maintain databases within an application area, working individually or coordinating database development as part of a team.

Work Role Identification: 621

Required Experience: Of the minimum seven years of required experience, at least five of those years must be specific to DoD systems, no specific field of experience is required other than what is identified in the function description above.

3.4.6.20 Software Developers, Advanced, 15-1252, Non-Key.

Function: Research, design, and develop computer and network software or specialized utility programs. Analyze user needs and develop software solutions, applying principles and techniques of computer science, engineering, and mathematical analysis. Update software or enhance existing software capabilities. May work with computer hardware engineers to integrate hardware and software systems, and develop specifications and performance requirements. May maintain databases within an application area, working individually or coordinating database development as part of a team.

Work Role Identification: 621

Required Experience: Of the minimum seven years of required experience, at least two of those years must be specific to DoD systems, no specific field of experience is required other than what is identified in the function description above.

3.4.6.21 Software Quality Assurance Analysts and Testers, Intermediate, 15-1253.

Function: Develop and execute software tests to identify software problems and their causes. Test system modifications to prepare for implementation. Document software and application defects using a “bug” tracking system and report defects to software or web developers. Create and maintain databases of known defects. May participate in software design reviews to provide input on functional requirements, operational characteristics, product designs, and schedules.

Work Role Identification: 622

Required Experience: Of the minimum three years of required experience, at least two of those years must be specific to DoD systems.

3.4.6.22 Software Quality Assurance Analysts and Testers, Advanced, 15-1253.

Function: Develop and execute software tests to identify software problems and their causes. Test system modifications to prepare for implementation. Document software and application defects using a “bug” tracking system and report defects to software or web developers. Create and maintain databases of known defects. May participate in software design reviews to provide input on functional requirements, operational characteristics, product designs, and schedules.

Work Role Identification: 622

Required Experience: Of the minimum seven years of required experience, at least two of those years must be specific to DoD systems.

3.4.6.23 Web and Digital Interface Designers, Intermediate, 15-1255.

Function: Design digital user interfaces or websites. Develop and test layouts, interfaces, functionality, and navigation menus to ensure compatibility and usability across browsers or devices. May use web framework applications as well as client-side code and processes. May evaluate web design following web and accessibility standards and may analyze web use metrics and optimize websites for marketability and search engine ranking. May design and test interfaces that facilitate the human-computer interaction and maximize the usability of digital devices, websites, and software with a focus on aesthetics and design. May create graphics used in websites and manage website content and links.

Work Role Identification: 621

Required Experience: Of the minimum three years of required experience, at least two of those years must be specific to DoD systems.

3.4.7 Service Contract Act (SCA) Labor Qualifications.

The following lists the SCA labor categories required for this effort:

LABOR CATEGORY	BLS SOC CODE	SCA CODE
Computer Based Training Specialist/Instructor	13-1151	15050
Computer Systems Analyst II	15-1211	14102
Computer Systems Analyst III	15-1211	14103
Drafter/CAD Operator I	17-3012	30061
Drafter/CAD Operator II	17-3012	30062
Drafter/CAD Operator III	17-3012	30063
Drafter/CAD Operator IV	17-3012	30064
Electronics Technician Maintenance II	17-3024	23182
Electronics Technician Maintenance III	17-3024	23183
Engineering Technician II	17-3023	30082
Engineering Technician III	17-3023	30083
Engineering Technician IV	17-3023	30084
Engineering Technician V	17-3023	30085
Order Clerk II	43-3061	01192